Before the Lab Acts: Benchmarking Language Models for Constrained Compound Selection

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Abstract

Language models are increasingly connected to scientific tools and automated laboratories, but many evaluations stop at predictions, prose, or unconstrained plans. We introduce SpecGuard-Chem, an action-level benchmark for the point at which assay evidence becomes an experimental commitment. On 91 CARA-derived lead-optimization tasks, a system receives 50 measured support compounds, a fixed candidate pool, explicit eligibility rules, and a budget of 10 assays, then must issue an ordered, executable batch of candidate identifiers. Candidate outcomes are physically separated into a hash-bound scorer artifact. The benchmark evaluates scientific shortlist utility and regret independently from action validity, deterministic repair, latency, and cost. Corrected deterministic evidence spans rules-only utility 66.922, random-valid 68.469, similarity 73.288, the best assay-local QSAR model 74.966, and a hidden-outcome oracle 79.563, demonstrating nontrivial headroom beyond eligibility filtering. The preregistered model experiment compares basic and descriptor-enriched interfaces for three current provider conditions, with post-hoc repair applied to the same raw outputs rather than purchased as separate calls. A fixed one-card provider pilot has completed as an execution and provenance check but is excluded from model comparison. Provider results are intentionally withheld until the separately authorized 91-card run is complete. SpecGuard-Chem does not claim a new biological dataset or a complete model of drug discovery. It contributes a reproducible action-evaluation layer over CARA for testing whether an LLM adds value before it is trusted to spend laboratory resources.

Keywords: AI for science; automated laboratories; compound prioritization; lead optimization; large language models; CARA; QSAR; benchmarking.

1 Introduction

Automated laboratories couple computational experiment selection to physical execution. Robot-scientist and self-driving-laboratory systems have already integrated model-guided screening, chemical optimization, and robotic experimentation [14, 4]. More recently, LLM-based systems have planned chemical tasks, called expert tools, and controlled laboratory interfaces [2, 3]. As the distance between generated text and physical action narrows, evaluation must ask more than whether an answer sounds plausible: would the exact issued action be executable, and is it a good use of a finite experimental budget?

This paper isolates one decision primitive from a future drug-discovery loop: choosing the next assay batch from a known pool using sparse project-local activity evidence. The boundary is deliberately narrow. It excludes compound generation, synthesis planning, multi-assay optimization, and sequential learning. In return, every available action, constraint, model input, outcome, and opportunity cost can be audited exactly, and LLM-backed systems can be compared with transparent molecular rankers on identical information.

The activity-data substrate is CARA, which organizes experimentally measured compound activities into assay-local virtual-screening and lead-optimization tasks [11, 12]. At the inference level, our task is candidly CARA- style activity estimation plus eligibility filtering and top-k allocation. We do not claim a new raw dataset or an unrelated prediction problem. The new unit of evaluation is the action: exactly k valid identifiers whose joint utility and regret determine how well a finite laboratory budget was allocated.

We ask three questions:

1. **Decision utility:** do LLM-backed selectors issue higher-utility assay batches than random-valid, heuristic, similarity, and strong per-assay QSAR systems?
2. **Interface and representation:** does descriptor-enriched molecular evidence change shortlist utility relative to a basic interface?
3. **Executability and operational value:** how often are raw LLM actions valid, what does deterministic repair change when applied to the same response, and what latency/token/dollar cost buys any gain?

Decision utility is the paper's primary scientific question. Validity is an independent property: a compliant weak list is executable but unhelpful, while a promising invalid list cannot be run as issued. The benchmark therefore does not treat guardrail gains as scientific gains.

2 Positioning: from prediction to action

Compound-activity prediction benchmarks typically score per-compound estimates or rankings. CARA improves the realism of that evaluation by retaining assay- based tasks, sparse labels, and lead-optimization versus virtual-screening settings [11]. SpecGuard-Chem retains that substrate and adds an action contract and systems protocol at the point where predictions would be converted into a laboratory allocation.

Table 1: CARA is the data and task substrate; SpecGuard-Chem is an action- evaluation layer. The distinction is the evaluated output and its executable semantics, not a claim of unrelated molecular learning.

	CARA activity view	**SpecGuard-Chem action view**
System output	Query activity predictions or ranking	Ordered batch of exactly k candidate IDs
Evaluation unit	Compound/task predictive performance	Whole experimental allocation per task
Budget	Not intrinsic to each prediction	Explicit finite assay capacity
Constraints	Molecular prediction domain	Candidate eligibility plus executable output contract
Failure semantics	Prediction error	Opportunity cost and/or unexecutable action
Artifact protocol	Activity benchmark data/splits	Separate system inputs and hash-bound scorer outcomes; exact requests and eventual traces, costs, repair attribution

Chemistry-agent work has shown that tools and laboratory APIs can extend LLM capabilities [2, 3]. That makes interface quality and output reliability important, but neither demonstrates that an LLM should replace a task-specific molecular model. Our comparator ladder is therefore part of the contribution: an LLM result is meaningful only relative to small, reproducible rankers that use the same support evidence.

3 Benchmark construction

3.1 Corrected CARA import

The release uses CARA v1.0.1 and the official `LO_All` table and split [12]. Split JSON files associate each task key with integer row positions. The corrected importer resolves each integer positionally, then verifies that the source row's task identifier equals the split key. Out-of-range positions, task mismatches, duplicate candidate identities, and support–candidate identity overlap are hard errors.

An exhaustive audit resolves all 24,588 references: 5,000 support and 19,588 query rows across 100 tasks. The previous label-based importer was incorrect; all historical paper-50 cards and downstream results are retired rather than treated as an earlier benchmark version.

3.2 Decision cards and biological context

For each eligible task, the public artifact contains the CARA task, target, and endpoint identifiers; 50 support structures with measured activity; every candidate ID, structure, and permitted descriptor; constraints; and budget $k = 10$. Candidate activity is stored only in a separate evaluator artifact. Each outcome row is bound to the canonical hash of its public card, preventing labels from being silently paired with a changed task.

The measured quantity is CARA's supplied *pChEMBL Value*, a standardized negative-log molar activity scale on which higher values indicate greater potency. Every card and request states this scale and direction explicitly; endpoint labels such as IC50, Ki, EC50, and Potency retain the assay context but do not change the higher-is-better interpretation of the supplied pChEMBL value.

Of 100 official tasks, 91 contain at least ten feasible candidates. The nine excluded tasks have feasible counts 0, 0, 0, 0, 0, 0, 0, 6, and 8; every exclusion is recorded before model evaluation. No hidden outcome is used to select among eligible tasks.

Table 2: Frozen v0.1.0 composition. Each task has a distinct CARA target identifier.

Property	**Value**
Official CARA tasks considered	100
Included / excluded tasks	91 / 9
Support compounds per task	50
Total public candidates	18,205
Candidate-pool range / mean	52–967 / 200.055
Feasible-pool range / mean	12–579 / 110.165
Endpoint tasks	68 IC50; 17 Ki; 5 EC50; 1 Potency
Assay budget	10

Target and endpoint identifiers provide minimal biological context and enable task-coherence checks. They do not supply mechanism, selectivity, phenotypic context, ADMET, toxicity, or a multi-objective project profile. This is an assay-local potency decision benchmark, not a broad biological-reasoning test.

3.3 Action contract and constraints

For task c, let E_c be the visible support evidence, P_c the candidate pool, $V_c \subseteq P_c$ the candidates satisfying molecular eligibility, and $k = 10$. A system m emits a schema-conforming action for task c containing an ordered sequence $S_m(c)$ of candidate IDs. The action is valid exactly when the task identity matches, it has length k, it contains unique members of P_c, it excludes support identities, and every selected candidate lies in V_c.

Version 0.1.0 candidate eligibility requires parseable structures, molecular weight at most 500, and cLogP at most 4.5. The configured forbidden-substructure list is empty and is reported as such. These are simple, explicit eligibility rules; they are not substitutes for synthesis, medicinal-chemistry, or safety review.

Wider future loop (not evaluated)

design ⟶ make ⟶ test ⟶ analyse ⟶ update

SpecGuard-Chem action boundary

support evidence + finite pool	⟶	filter, estimate, rank	⟶	ordered top-10 assay batch
constraints + budget	⟶	system under test	⟶	hidden retrospective outcomes

Figure 1: The benchmark isolates one transition at which a model-generated ranking becomes a finite experimental commitment. The surrounding autonomous laboratory loop is motivation, not an evaluated claim.

4 Systems and interfaces

4.1 Non-language comparators

We include a random-valid selector; a descriptor-desirability heuristic; similarity to the most active support compound using Morgan fingerprints [10]; and per-task random-forest, gradient-boosting, and linear support-vector regressors trained only on the 50 visible support measurements [9, 13]. A non-deployable oracle ranks feasible candidates using hidden outcomes and defines attainable headroom. Cheminformatic parsing, descriptors, and fingerprints use RDKit [7].

4.2 LLM interfaces and frozen conditions

The minimum paper matrix crosses 91 cards, two interfaces, and three model conditions (546 raw provider requests). The *basic* interface supplies the action contract, assay context, support evidence, candidate IDs, structures, molecular weight, and cLogP. The *descriptor-enriched* interface adds precomputed TPSA, hydrogen-bond donor/acceptor counts, and rotatable bonds. It does not add hidden activities or external search.

The frozen paper conditions are a dated GPT-5.5 snapshot, Claude Opus 4.8, and DeepSeek V4 Pro as checked on 16 July 2026 [8, 1, 5]. Exact requests are frozen before execution. Provider-returned model identifiers, settings, timestamps, token counts, latency, and errors are retained in traces. A conservative estimate places the full uncached matrix at $106.06, with an explicit $120 stop gate. No call occurs unless the command also receives `-allow-external`.

4.3 Post-hoc deterministic repair

Raw behavior and guarded-system behavior are evaluated from the same provider response. The validator checks JSON shape, task identity, selection count, candidate membership, duplicates, support exclusion, and candidate constraints; it never reads activity outcomes. When repair is requested, valid raw IDs are retained in order and missing slots are filled by the deterministic rules-only ranking. The raw response, raw issues, repaired output, and repair operations remain attributable. This design avoids conflating a different repair prompt and an additional paid model call with deterministic harness behavior.

5 Evaluation

5.1 Utility, ranking, and regret

Let y_{ic} be the hidden pChEMBL activity for candidate i in task c, with higher values better. Feasible utility is

$$U_m(c) = \sum_{i \in S_m(c)} y_{ic} \, \mathbf{1}\{i \in V_c\}. \tag{1}$$

Invalid selections receive no utility. NDCG@k measures ordered ranking quality using hidden activity as graded relevance [6]. Constrained regret is

$$R_m(c) = \sum_{i \in O_c} y_{ic} - U_m(c), \tag{2}$$

where O_c is the feasible hidden-outcome top-k. Lower regret is better.

5.2 Action validity and operational outcomes

Whole-action validity is a binary card-level measure: it equals one only when the complete output has no contract issue, including schema shape, task identity, exact size, in-pool membership, uniqueness, support exclusion, and candidate eligibility. We separately report the valid-selection fraction (`compliance_rate`), defined as valid selected entries divided by k, as a partial-credit diagnostic rather than a synonym for action validity. For example, ten feasible selections returned under the wrong task identifier have selection compliance one but action validity zero. For every LLM interface we report raw metrics and the corresponding post-hoc-repaired metrics, plus repair rate and violation taxonomy. Latency, input/output tokens, cache status, and estimated or actual cost are reported per condition.

Means and bootstrap intervals are computed across the same 91 task cards. Primary comparisons use paired card-level resampling and report the LLM-minus- best-QSAR utility/regret difference, not only independent confidence intervals.

6 Results

6.1 The corrected benchmark separates molecular selectors

Table 3 establishes the benchmark's useful dynamic range before provider spending. Eligibility filtering alone is not sufficient: rules-only utility is 66.922 and random-valid utility is 68.469. Similarity rises to 73.288, while all three assay-local QSAR models reach approximately 74.75–74.97. The oracle remains at 79.563, leaving 4.596 utility points of mean headroom above the best deployable baseline. All rows have perfect action validity, so their separation is attributable to ranking rather than malformed actions.

Table 3: Corrected deterministic results on all 91 cards. Utility and NDCG bootstrap intervals are task-level 95% percentile intervals. Oracle is not a deployable system.

System	Utility (95% CI)	NDCG@10	Regret	Action validity
Oracle valid top-k	79.563 (77.486–81.812)	1.000	0.000	1.000
QSAR linear SVR	74.966 (72.830–77.032)	0.938	4.596	1.000
QSAR random forest	74.958 (72.825–77.142)	0.938	4.605	1.000
QSAR gradient boosting	74.750 (72.621–76.859)	0.935	4.813	1.000
Similarity to best active	73.288 (71.264–75.236)	0.919	6.274	1.000
Random valid	68.469 (66.487–70.450)	0.855	11.094	1.000
Rules/desirability	66.922 (64.772–69.062)	0.828	12.641	1.000

6.2 Do LLMs add decision value?

This subsection is a declared pre-run placeholder. The 546 exact requests and conservative cost estimate are frozen. Six responses for one fixed card have been acquired and audited solely to verify execution, provenance, scoring, and raw-versus-repaired attribution; they are not a model comparison and cannot open the paper's 91-card result gate. Consequently, this manuscript does not reuse or report any historical paper-50 LLM number. The final paper will lead with paired per-task utility and regret versus QSAR, followed by representation, raw validity, post-hoc repair, latency, tokens, and cost.

7 Discussion

7.1 A benchmark with a deliberately simple core

The formal core of this task is filter–predict–rank–allocate. This simplicity does not establish a new molecular-learning problem, and the paper should not pretend otherwise. It does create a precise evaluation boundary at the point where a model's scores become an experimental commitment. That boundary makes invalid actions, opportunity cost, deterministic repair, system inputs, and operational cost measurable together.

The corrected baseline ladder shows that the benchmark has independent legs: constraints do not collapse the task, transparent molecular models improve materially on heuristic selection, and oracle headroom remains. If LLMs fail to beat QSAR, the result is still informative for system design: a future lab should route this action to the smaller specialized ranker. If an LLM improves on QSAR, the paired per-task analysis and interface ablation identify where that incremental value appears.

7.2 What the benchmark does not establish

Retrospective activity is not prospective validation. CARA inherits ChEMBL measurement and curation heterogeneity [15]. The objective is assay-local potency, not selectivity, mechanism, ADMET, toxicity, synthesis feasibility, or therapeutic value. The candidate pool is fixed; no structure is generated. Simple molecular thresholds are eligibility rules, not medicinal- chemistry judgment. One-shot allocation does not test learning across rounds, and snapshot-specific provider results will not generalize automatically to future models.

These limits are reasons to keep the claim narrow, not reasons to postpone the release indefinitely. A verified action-level unit test is a useful precursor to richer biological and sequential extensions, provided the paper never equates success here with autonomous-laboratory readiness.

8 Reproducibility and release

The planned archival release will separate public cards from scorer outcomes and include source and transform provenance, exclusions, exact requests, model configurations, traces, per-card scores, aggregate tables, costs, environment locks, licenses, citation metadata, a file manifest, and SHA256 checksums. Historical invalid result tags will remain immutable but are superseded explicitly. The semantic release tag will be created only after a clean-checkout build verifies the bundle and compiled manuscript.

9 Conclusion

Before giving an LLM an entire automated laboratory loop, we should test one consequential action against strong non-language alternatives under a contract that can be audited exactly. SpecGuard-Chem turns CARA's assay-grounded evidence into such a test: a finite, constrained next-batch decision with hidden outcomes, measurable opportunity cost, and explicit executability. Its claim is modest but operationally important—whether an LLM adds value at the moment a prediction would begin to spend laboratory resources.

Data, code, and AI-use statement

Release artifacts and exact reproduction commands will be bundled with the tagged repository. Code is MIT licensed; CARA and CARA-derived data retain CC BY 4.0 terms. ChatGPT/Codex assisted with project framing, code and artifact auditing, manuscript structuring/editing, and LaTeX compilation. All numerical claims are checked against local machine-readable artifacts. No patient or clinical data are used. Author list, contributions, funding, competing interests, and venue- specific disclosure wording require human confirmation before submission.

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